

Q23 If slope of line passing through point $(\alpha, 6)$ and $(-2, 5)$ is $\frac{1}{3}$, then $\alpha =$ — (A) -1 (B) 1 (C) 5 (D) 3

Solⁿ

$$\frac{1}{3} = \frac{5-6}{-2-\alpha} \Rightarrow \frac{1}{3} = \frac{-1}{-2-\alpha} \Rightarrow \frac{1}{3}(-2-\alpha) = -1$$

$$-2-\alpha = -3$$

$$-\alpha = -3+2$$

$$-\alpha = -1$$

$$\alpha = 1$$

Q24 Slope of line making an angle of measure $\frac{\pi}{4}$ with the positive direction of X axis is — (A) 1 (B) $\frac{1}{\sqrt{3}}$ (C) $\sqrt{3}$ (D) ∞

Solⁿ

Q25 If slope of line is $\sqrt{3}$, then its angle with X axis is — (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Solⁿ

Q26 slope of line $2x-3y+5=0$ is — (A) $-\frac{2}{3}$ (B) $-\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

Solⁿ

Q27 Slope of line $(\cos \alpha)x + (\sin \alpha)y = 5$ is — (A) $\tan \alpha$ (B) $\cot \alpha$ (C) $-\tan \alpha$ (D) $-\cot \alpha$

Solⁿ

Q28 Slope of line $x-3=0$ is — (A) 0 (B) $\frac{\pi}{2}$ (C) ∞ (D) 1

Solⁿ

Q29 Slope of line $y-3=0$ is — (A) 0 (B) $\frac{\pi}{2}$ (C) ∞ (D) 3

Solⁿ

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⑧ Equation of line (2-point form, slope point form, Intercept form, General form) :- (24)

<i> 2-point form :- $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

<ii> slope-point form :- $y - y_0 = m(x - x_0)$

where (x_0, y_0) = given point, m = slope

<iii> Intercept form :- $\frac{x}{a} + \frac{y}{b} = 1$ or $ax + by + c = 0$
 where $a = x$ intercept
 $b = y$ intercept
 x intercept = $-\frac{c}{a}$
 y intercept = $-\frac{c}{b}$

④ <iv> General Form :- $Ax + By + C = 0$

<i> Ex: 10 find Equⁿ of line passing from (2, 3) & (3, -1).
 Solⁿ

<ii>

<i> Ex: 11 Find equⁿ of line passing from (1, 2) & (2, 1)
 Solⁿ

Ex:

S Ex: 12 Find equⁿ of line whose slope is 3 & passing from (1, 2).
 Solⁿ

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Ex 13 X intercept of line $3x+5=0$ is —
 (A) $-5/3$ (B) $5/3$ (C) $3/5$ (D) 0
 Solⁿ

Ex 14 X intercept of line $x \cos \alpha + y \sin \alpha = 2$ is —
 (A) $2 \cos \alpha$ (B) $2 \sec \alpha$ (C) $-2 \cos \alpha$ (D) $-2 \sec \alpha$
 Solⁿ

Ex 15 Y intercept of line $4x+3y+2=0$ is —
 (A) $-2/3$ (B) $-2/4$ (C) $-1/3$ (D) None
 Solⁿ

(C) Condition of Parallel & Perpendicular lines :-
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Parallel line : Slopes are equal ($m_1 = m_2$)
 Perpendicular line : $m_1 \times m_2 = -1$

(D) Equation of Parallel & Perpendicular lines :-

→ General Form : $Ax + By + C = 0$
 Parallel line Equⁿ : $Ax + By + K = 0$
 Perpendicular line Equⁿ : $Bx - Ay + K = 0$

Ex 16 Slope of line perpendicular to line $5x-7y+3=0$ is —
 (A) $5/7$ (B) $7/5$ (C) $-7/5$ (D) $7/3$
 Solⁿ

Ex 17 Slope of line perpendicular to line $2x+3y=7$ is —
 (A) $2/3$ (B) $3/2$ (C) $-2/3$ (D) $-3/2$
 Solⁿ

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Ex 18 If Two lines having slopes m_1 and m_2 are Parallel to each other then — (A) $m_1 \cdot m_2 = -1$ (B) $m_1 \cdot m_2 = 1$ (C) $m_1 \cdot m_2 = 0$ (D) $m_1 = m_2$ (26)

Solⁿ

Ex 19 If Two lines having slopes m_1 and m_2 are parallel to each other then — (A) $m_1 \cdot m_2 = 1$ (B) $m_1 \cdot m_2 = -1$ (C) $m_1 \cdot m_2 = 0$ (D) $m_1 = m_2$

Solⁿ

Ex 20 The Eqⁿ of line Parallel to line $x + 2y + 2 = 0$ is — (A) $2x + y + 2 = 0$ (B) $2x - y + 2 = 0$ (C) $x + 2y + 3 = 0$ (D) $x - 2y + 2 = 0$

Solⁿ

Ex 21 The Equation of line Parallel to line $3x - 2y + 12 = 0$ is — (A) $3x - 2y - 12 = 0$ (B) $2x - 3y + 12 = 0$ (C) $3x + 2y + 12 = 0$ (D) $2x + 3y + 12 = 0$

Solⁿ

Ex 22 The Equation of line Perpendicular to line $4x + 3y - 12 = 0$ is — (A) $3x - 4y + k = 0$ (B) $3x + 4y + k = 0$ (C) $3x - 3y + k = 0$ (D) $4x - 4y + k = 0$

Solⁿ

(E) Angle between two lines: —
→ slope of line-1 is m_1 and slope of line-2 is m_2 then
Angle betⁿ two line: $\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Ex 23 Angle between line $x + y = 0$ & $x - y = 0$ is — (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) 0 (D) $\frac{\pi}{2}$

Solⁿ

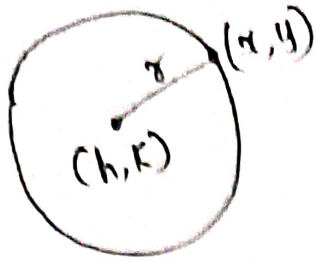
Ex 24 Angle of line $x - 5 = 0$ with positive direction X axis is — (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) 0

Solⁿ

{A 24 its 12 → 41702121}

Equation of Circle with Centre and Radius:-

(27)



$$(x-h)^2 + (y-k)^2 = r^2$$

Where (h, k) = Centre

r = radius

→ If $(h, k) = (0, 0)$ then $x^2 + y^2 = r^2$

(A) To find Centre and Radius from General form of Circle:-

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

→ Coefficient of x^2 and y^2 is Unique (1).

→ Compare it with given Eqn, You will get $g = \text{---}$
 $f = \text{---}$

$$\begin{aligned} \rightarrow \text{Centre } C(-g, -f) \\ \rightarrow \text{Radius } r = \sqrt{g^2 + f^2 - c} \end{aligned}$$

Ex-25 Centre of circle $x^2 + y^2 + 2gx + 4y + 1 = 0$ is —
(A) (1, 2) (B) (1, -2) (C) (-1, -2) (D) (-1, 2)

Soln

Ex-26 Centre of circle $x^2 + y^2 - 4x + 7 = 0$ is —
(A) (-2, 0) (B) (2, 0) (C) (-4, 0) (D) (0, 4)

Soln

Ex-27 Radius of circle $x^2 + y^2 - 2x + 4y + 1 = 0$ is —
(A) $\sqrt{6}$ (B) 2 (C) 4 (D) 6

Soln

Ex-28 Radius of circle $x^2 + y^2 = 25$ is — (A) -5 (B) 25 (C) 5 (D) ± 5

Soln

Ex-29 Radius of circle $x^2 + y^2 = 50$ is — (A) $5\sqrt{2}$ (B) $2\sqrt{5}$ (C) 5 (D) 10

Soln

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Ex 30 Radius of circle $x^2 + y^2 - 4x - 6y + 4 = 0$ is —
(A) 4 (B) 6 (C) 2 (D) 3

Solⁿ

Ex 31 Equation of line perpendicular to line $3x - 2y + 7 = 0$ and passes through (8, 2) is —
(A) $3x + 2y - 11 = 0$ (B) $2x + 3y - 5 = 0$
(C) $2x + 3y + 10 = 0$ (D) $2x + 3y - 12 = 0$

Solⁿ

Ex 32 Radius of circle $2x^2 + 2y^2 + 2x + 4y - 2 = 0$ is —
(A) 7 (B) $\sqrt{7}$ (C) $\frac{3}{2}$ (D) $\frac{3}{\sqrt{2}}$

Solⁿ

Ex 33 Radius of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is —
(A) $g^2 + f^2 - c$ (B) $\sqrt{g^2 + f^2 - c}$ (C) $\sqrt{g^2 + f^2 - c}$ (D) $\sqrt{g^2 + f^2 - c^2}$

Solⁿ

Ex 34 Radius of circle $x^2 + y^2 = 18$ is — (A) $2\sqrt{3}$ (B) $3\sqrt{2}$ (C) $2\sqrt{2}$ (D) $3\sqrt{3}$

Solⁿ

Ex 35 Radius of circle $x^2 + y^2 + gx - fy = 0$ is —
(A) $4\sqrt{g^2 + f^2}$ (B) $2\sqrt{g^2 + f^2}$ (C) $\frac{1}{2}\sqrt{g^2 + f^2}$ (D) $\frac{1}{4}\sqrt{g^2 + f^2}$

Solⁿ

Ex 36 If centre of circle $x^2 + y^2 + 3px - 1 = 0$ is (6, 0) then p = —
(A) 4 (B) -4 (C) 2 (D) -2

Solⁿ

— x — x —

ALL THE BEST
{Prepared By}
{Arpit Sir}
{41414141}